Kazakh-British Technical University
School of Information Technology and Engineering

SCADA systems and Industrial networks

THE COMMUNICATION PRINCIPLES OF SCADA CONTROLLERS

Laboratory work №13

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Almaty, 2026

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INTRODUCTION

In this laboratory work, I focused on the fundamental principles of integrating third-party controllers into the Honeywell Experion PKS system using SCADA interfaces. The primary objective was to understand how the Experion server establishes and manages communication with external devices via industrial protocols.

For this exercise, the Modbus TCP protocol was used as the primary communication standard. My tasks involved the step-by-step configuration of the data exchange environment, ranging from creating individual communication channels to performing bulk configurations of multiple interfaces using automated naming tools.

Throughout the process, I utilized key engineering software, specifically Configuration Studio for overall system navigation and Quick Builder for detailed SCADA database management. Special attention was paid to communication reliability parameters, such as Marginal and Fail alarm limits, as well as diagnostic scan rates. These settings are critical for ensuring stable monitoring and control within a Distributed Control System (DCS) environment.

1. Tasks

1.1 Task 1. Configuring a Channel

I began by launching the Configuration Studio application and establishing a connection to the system. In the left-hand navigation pane, I followed the hierarchy: [System Name] → Servers → [My Server Name]. From there, I selected the Control Strategy section and clicked once on the Build Channels option.

Inside the Quick Builder environment, I proceeded to create a new communication path. To do this, I either clicked the Add icon on the toolbar or right-clicked the Channels folder in the configuration tree and selected Add Item.

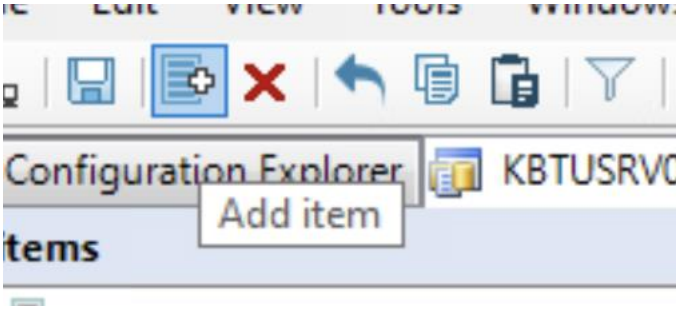

Figure 1 – "Add item" icon

For the purposes of this laboratory work, I located and selected the Modbus Channel type. I assigned a unique identifier to the new channel by typing M_CH1 in the Name field. After clicking OK, the channel was successfully added to the configuration.

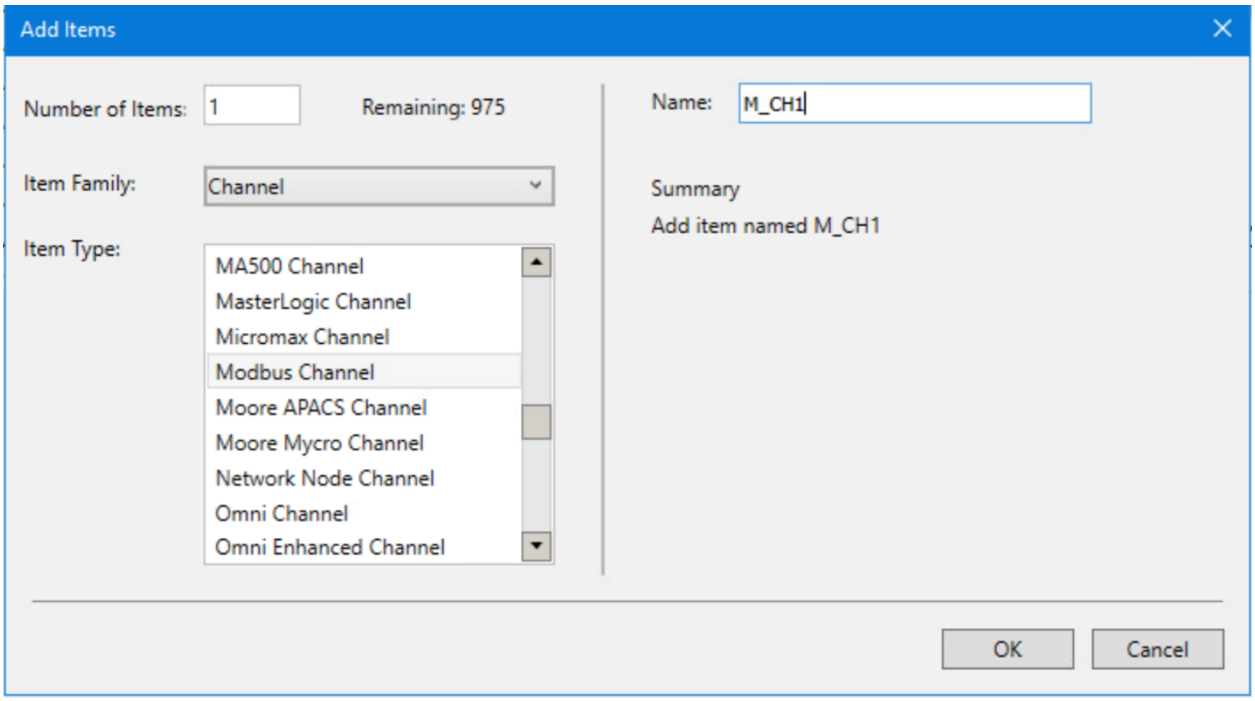

Figure 2 – Adding new channel M_CH1

On the Main tab of the channel details, I entered the specific operational parameters as defined in the technical requirements. I set the description to "Modbus Connection Channel" and associated it with the SysModel asset. Crucially, I configured the communication thresholds by setting the Marginal Alarm Limit to 20 and the Fail Alarm Limit to 40; these values determine how many failed connection attempts the system tolerates before triggering a warning or a failure status.

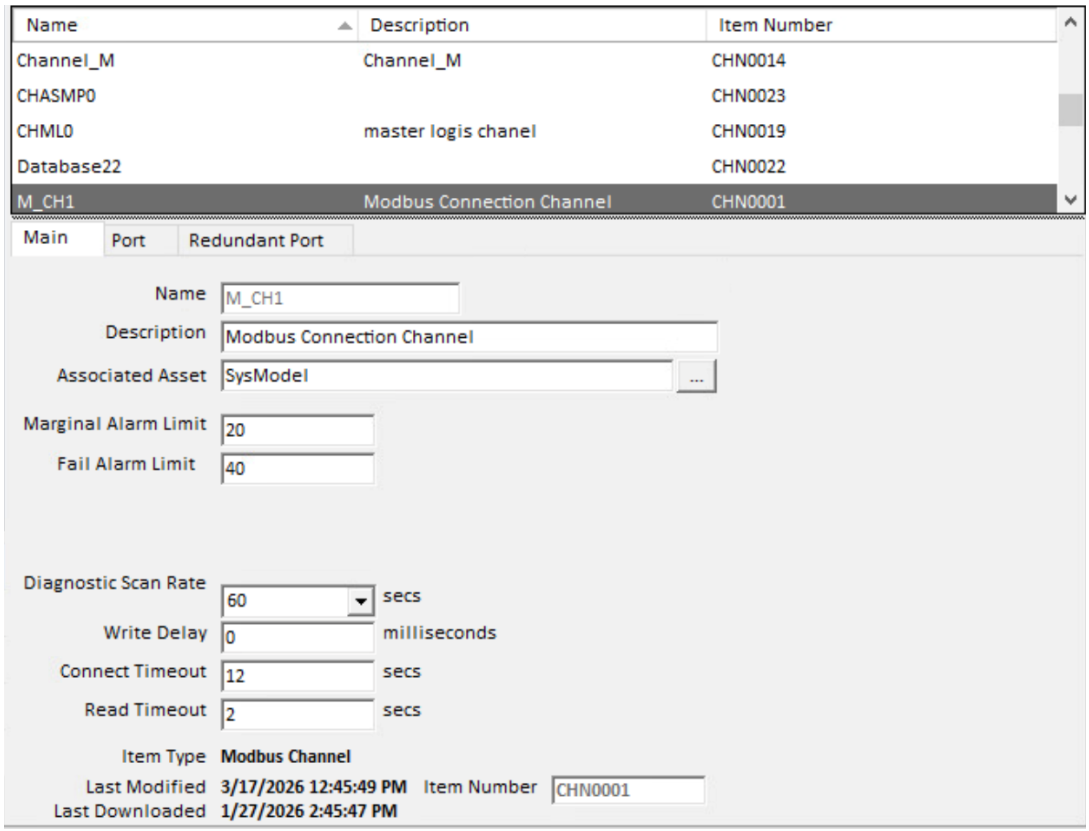

Figure 3 – M_CH1 channel configuration (Main tab)

Next, I navigated to the Port tab to establish the physical communication interface for the Modbus protocol. From the available options, I selected LANVendor as the Port Type and specified Modbus TCP as the Network Type.

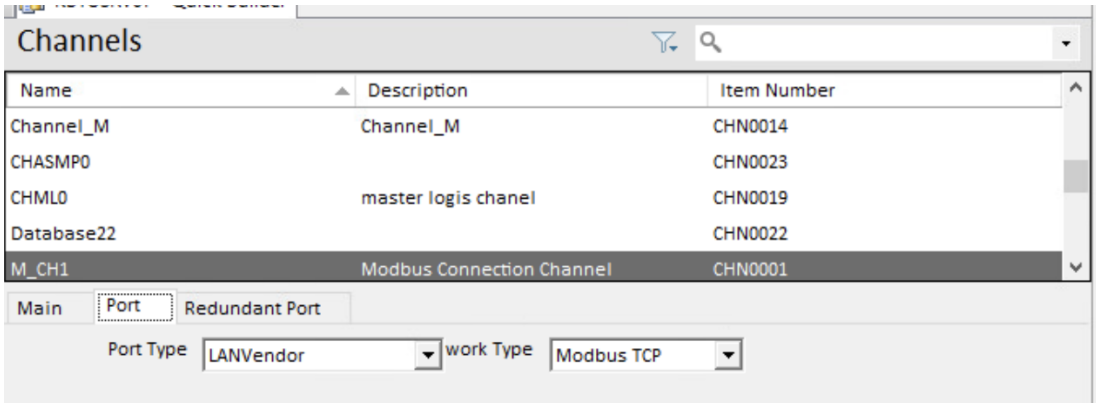

Figure 4 – M_CH1 channel configuration (Port tab)

I proceeded to the Redundant tab and verified that the Port Type was set to None, as a redundant communication path was not required for this specific setup.

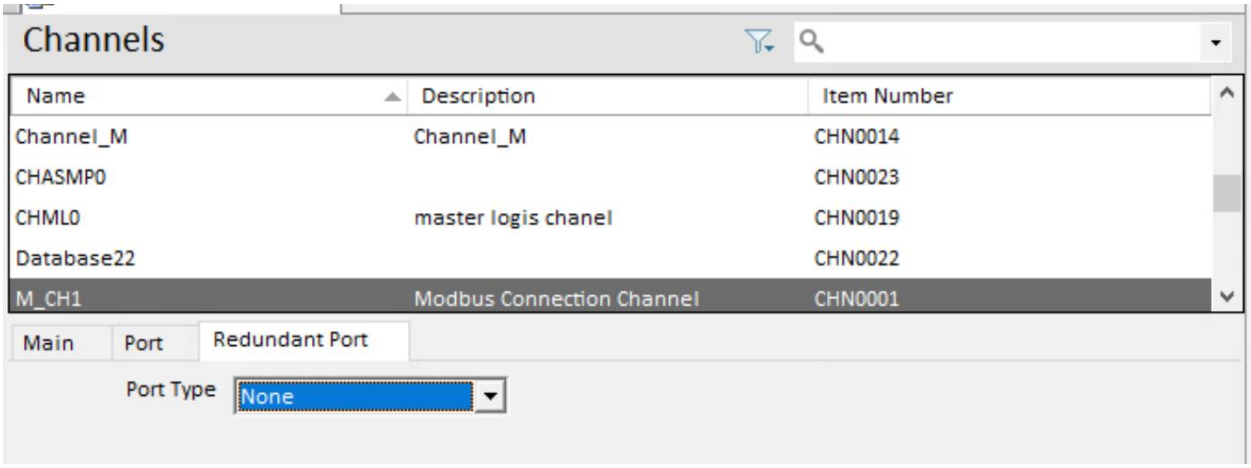

Figure 5 – M_CH1 channel configuration (Redundant Port tab)

Once the configuration was complete, I clicked the Download icon on the toolbar to begin the process of transferring these settings from the Quick Builder database to the active Server Runtime.

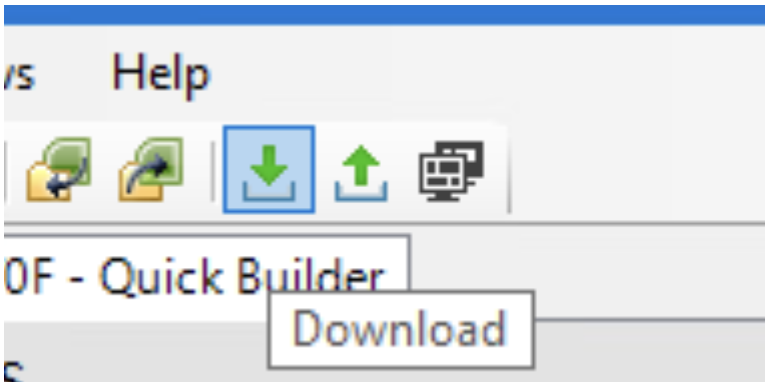

Figure 6 – Download icon

In the download window, I selected the Selected Items Only option and confirmed that exactly "1 Modbus Channel" was displayed in the Summary frame. I then accessed the Edit Options menu, where I enabled the Automatically download deletions feature while ensuring the Rebuild scan tables option remained disabled to avoid unnecessary overhead.

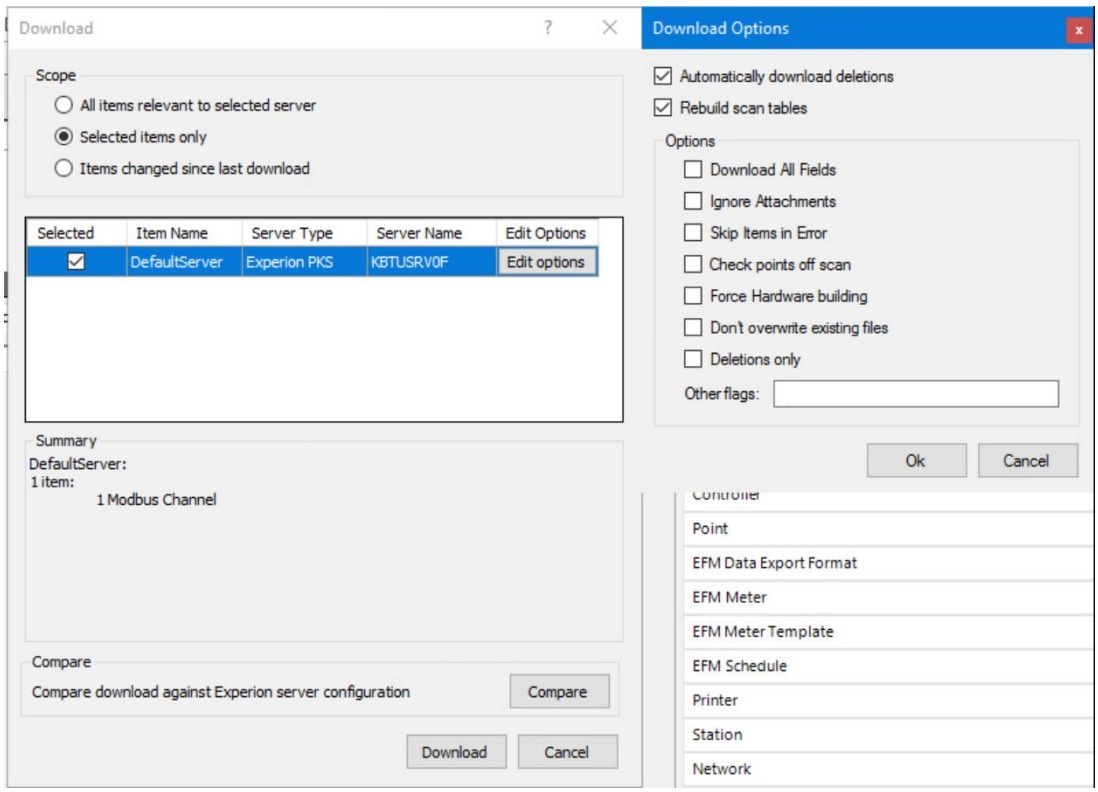

Figure 7 – Download window options

After clicking Download, I monitored the status to ensure the action was successful. If any errors had appeared, I was prepared to troubleshoot the configuration by reviewing the QBDB.out or Error.log files to identify the root cause.

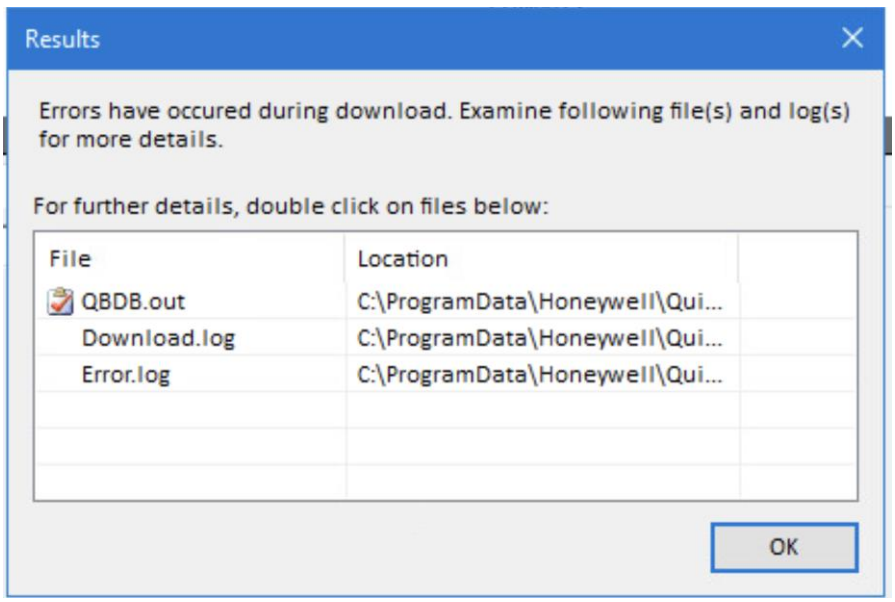

Figure 8 – After Download results

To verify that the newly configured channel was successfully integrated into the runtime environment, I navigated to the SCADA Controllers display. I followed the path: View → System Status → SCADA Controllers from the main menu.

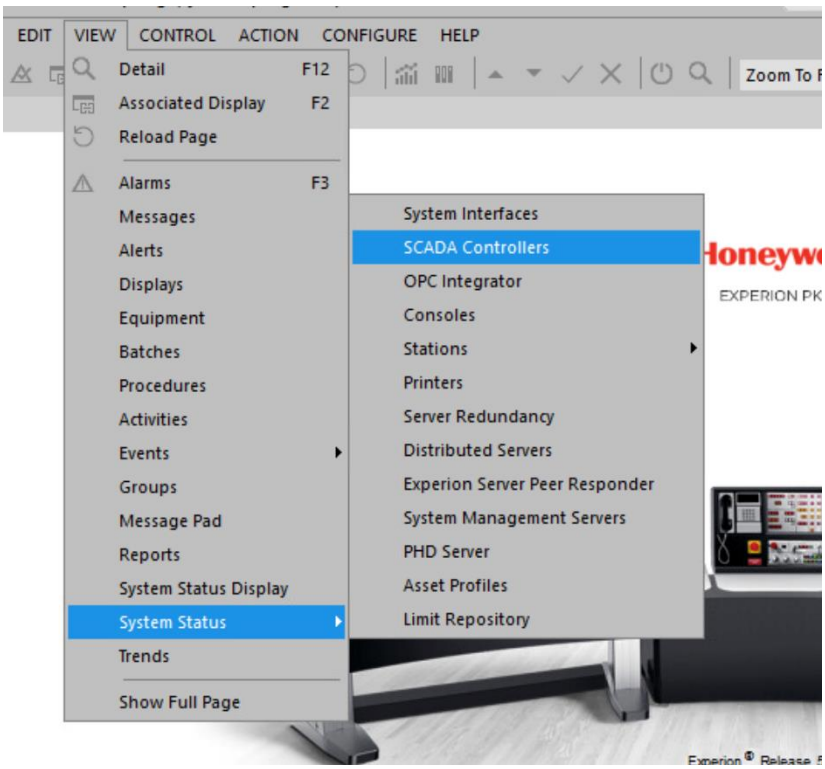

Figure 9 – Path to SCADA Controllers display

From the list of available channels in the SCADA Controllers display, I located my channel, M_CH1. I double-clicked on this entry to open the specific SCADA Channel detail window. This window allows for direct interaction with the channel's runtime state, which is separate from the offline configuration I performed in Quick Builder.

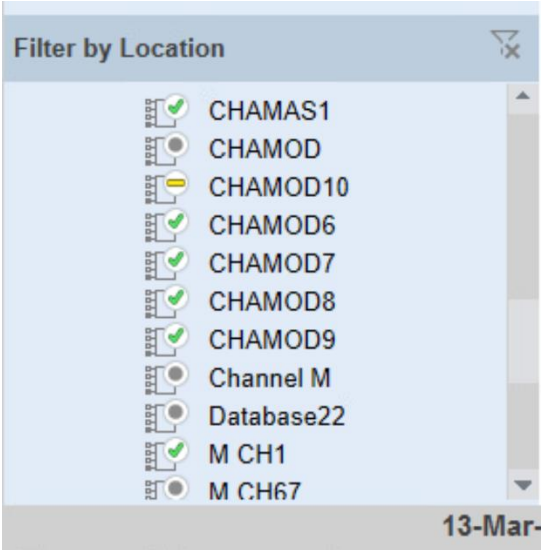

Figure 10 – Available channels

In the SCADA Channel window, I located the Enable checkbox. Upon selecting this box, the system initiated the communication sequence for the Modbus protocol. I monitored the status indicator and confirmed that the channel state transitioned to OK. This successful status change indicated that the server was correctly attempting

to communicate and that the channel was now active and ready to support controller connections.

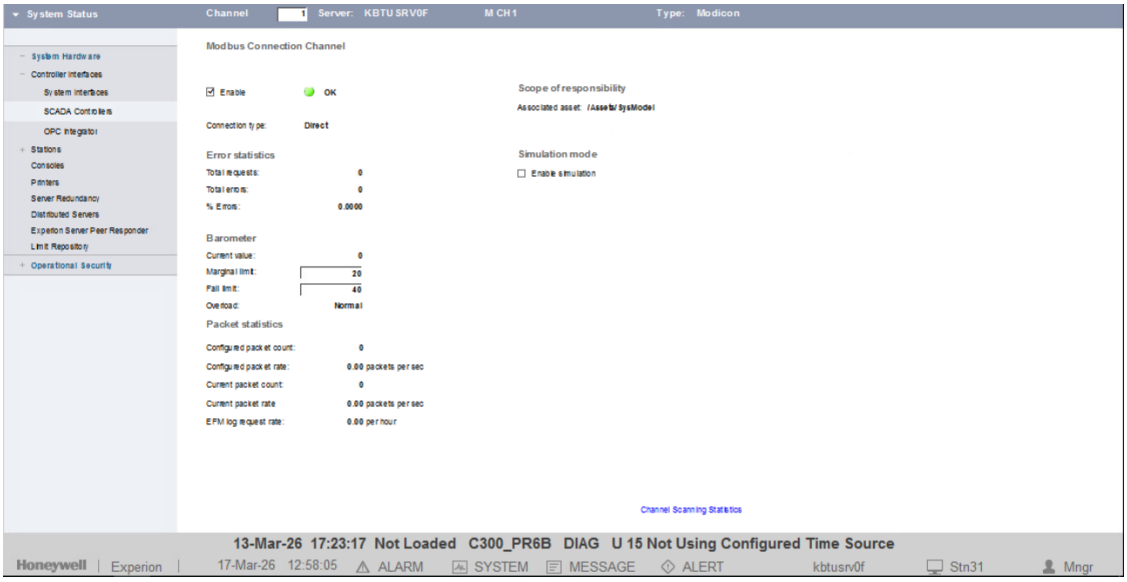

Figure 11 – SCADA Channel detail window.

To streamline the configuration process for multiple interfaces, I utilized the bulk addition feature in Quick Builder. I initiated a new "Add Item" command and configured the generation settings to create five new channels simultaneously. Following the requirements in Table 2, I set the Number of Items to 5 and selected Modbus Channel as the network type. To ensure consistent naming, I defined the Prefix as "M_CH", set the Variable to Numeric with 1 digit starting from 5, and added the Suffix "_SCADA". This resulted in the automatic generation of channels named M_CH5_SCADA through M_CH9_SCADA.

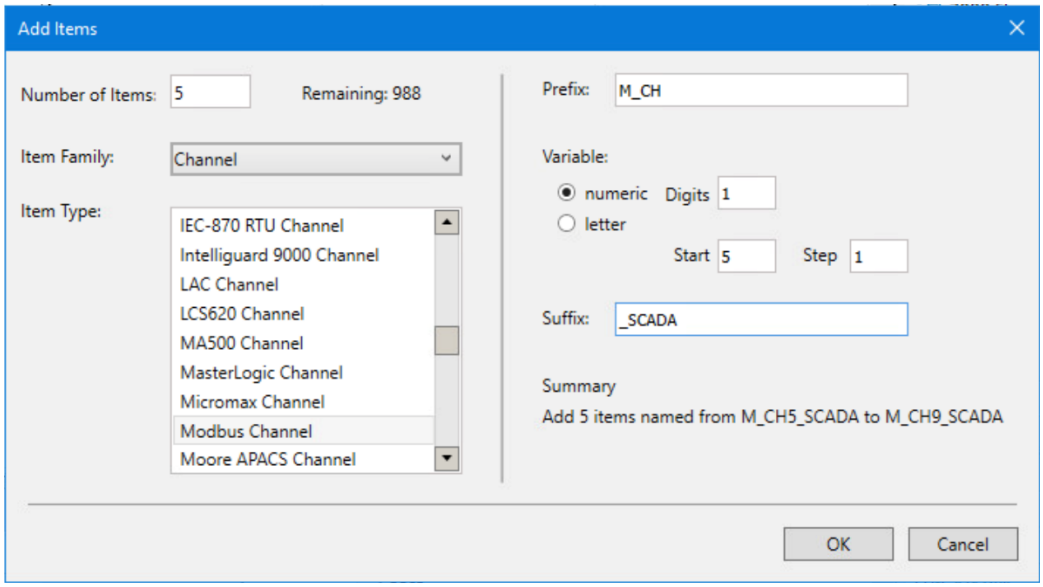

Figure 12 – Adding 5 new channels

After the channels were generated, I systematically applied the operational settings to each of the five new entities (M_CH5_SCADA to M_CH9_SCADA). For each channel, I repeated the configuration of the Main and Port tabs.

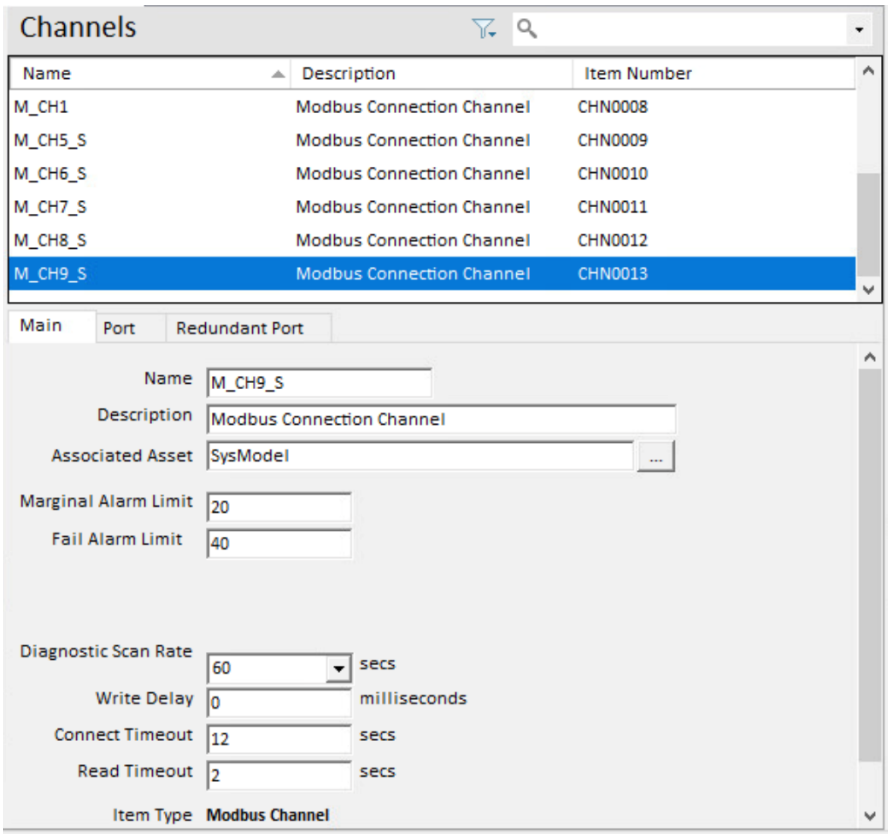

Figure 13 – New channel configurations

Once all parameters were validated, I initiated the Download process. In the download dialog, I ensured that all five new Modbus channels were listed in the summary frame.

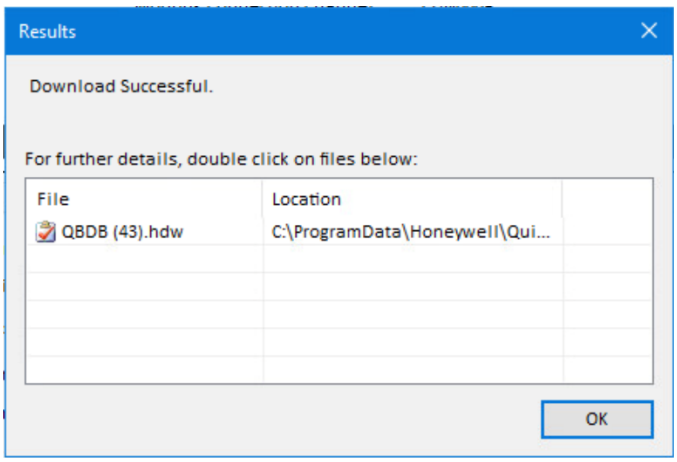

Figure 14 – After Download results

CONCLUSION

In this laboratory work, I successfully explored the principles of SCADA controller communication within the Honeywell Experion PKS environment. The focus was on the practical application of the Modbus TCP protocol, which serves as an industry standard for integrating third-party controllers. The configuration process covered all essential stages, from building the logical channel structure in Quick Builder to activating them in real-time via the Station interface.

A significant aspect of this work was mastering bulk configuration tools. I learned how to use the "Number of Items" and naming convention features to efficiently generate a group of channels (from M_CH5_SCADA to M_CH9_SCADA). This skill is crucial for large-scale industrial projects where configuring hundreds of communication nodes manually would be impractical.

Furthermore, I studied the mechanisms used to ensure data transmission reliability. By configuring the Marginal and Fail Alarm limits, I gained an understanding of how the system diagnostics monitor network health and notify operators of communication degradation or total failure. The successful download of the configuration and the subsequent "OK" status in the SCADA Controllers display confirmed that the settings were correct and the system was ready for live data exchange.

WORKSHEET

Student name: Fazkullin Farid, Zhanshin Temirlan

1. **(50 points)** Find out information about RS-485, RS-422 and RS-232.

Interface	Features	Bandwidth and connection distance
RS-485	Multi-point (up to 32 nodes), differential signaling, high noise immunity. Ideal for industrial networks.	Distance: up to 1200 m. Speed: 100 kbps – 2.5 Mbps.
RS-422	Single-master, multi-drop (up to 10 nodes), differential signaling. Provides balanced data transmission.	Distance: up to 1200m. Speed: up to 2.5 Mbps.
RS-232	Point-to-point (only 2 devices), single-ended signaling. Common for simple PC-to-controller connections.	Distance: up to 15 m. Speed: up to 38.4 kbps.

2. **(20 points)** What does mean LANvendor port type?

The LANvendor port type in Honeywell Experion PKS refers to a logical communication interface used when the SCADA channel communicates with a controller over an Ethernet network (Local Area Network) rather than a direct serial connection.

It acts as a driver or a software "gateway" that allows the server to use the computer's Network Interface Card (NIC) to send and receive industrial protocol packets (such as Modbus TCP) to devices connected via a standard network switch or router.

3. **(30 points)** What are the differences between the Network Interface and Network Protocol?

The difference between a Network Interface and a Network Protocol lies in the distinction between the hardware and the software layers: the Network Interface refers to the physical hardware or connection point (such as a Network Interface Card or Ethernet cable) that enables a device to join a network, whereas the Network Protocol is the set of logical rules and standards (such as Modbus TCP or TCP/IP) that determines how data is formatted, addressed, and transmitted. In essence, the interface provides the physical "pathway" for communication, while the protocol provides the "language" that allows devices to understand and process the exchanged information.